

Auctions

Tentative map of the auction literature

- An auction is any “institution” used to buy or sell goods or services.
- There are two main models, IPV and Affiliation, and two set of questions, properties of certain auctions and the search for an optimal mechanism.

	IPV	Affiliation
Ad-hoc auctions		
Optimal mechanism		

Auction examples

Auctions differ on (a) how the winner of the object is selected and (b) how much bidders pay.

- English Auction (EA). Single unit.
- Second-Price Sealed-Bid Auction (SPA). Single unit.
- First-Price Sealed-Bid Auction (FPA). Single unit.
- Dutch Auction (DA). Single unit.
- Uniform or discriminatory price auctions. Multi units.
- Multi-unit Vickrey Auction.
- Ausubel Auction. Multiple units.

FPA and SPA are examples of Bayesian games.

Variations within an auction

- Reserve prices
- Entry fee.
- Limited time for submitting bids.
- Payment contingent variables other than bids (royalties).
- Minimum accepted increment to standing bid.
- Sale of shares of the object instead of the unity.

Variations

- Variations in models: (non)symmetry, risk-aversion, collusion, budget-constrained bidders, single vs multi-unit auctions, (non) independence, etc.
- Variations in objective: efficiency vs seller's revenue, etc.

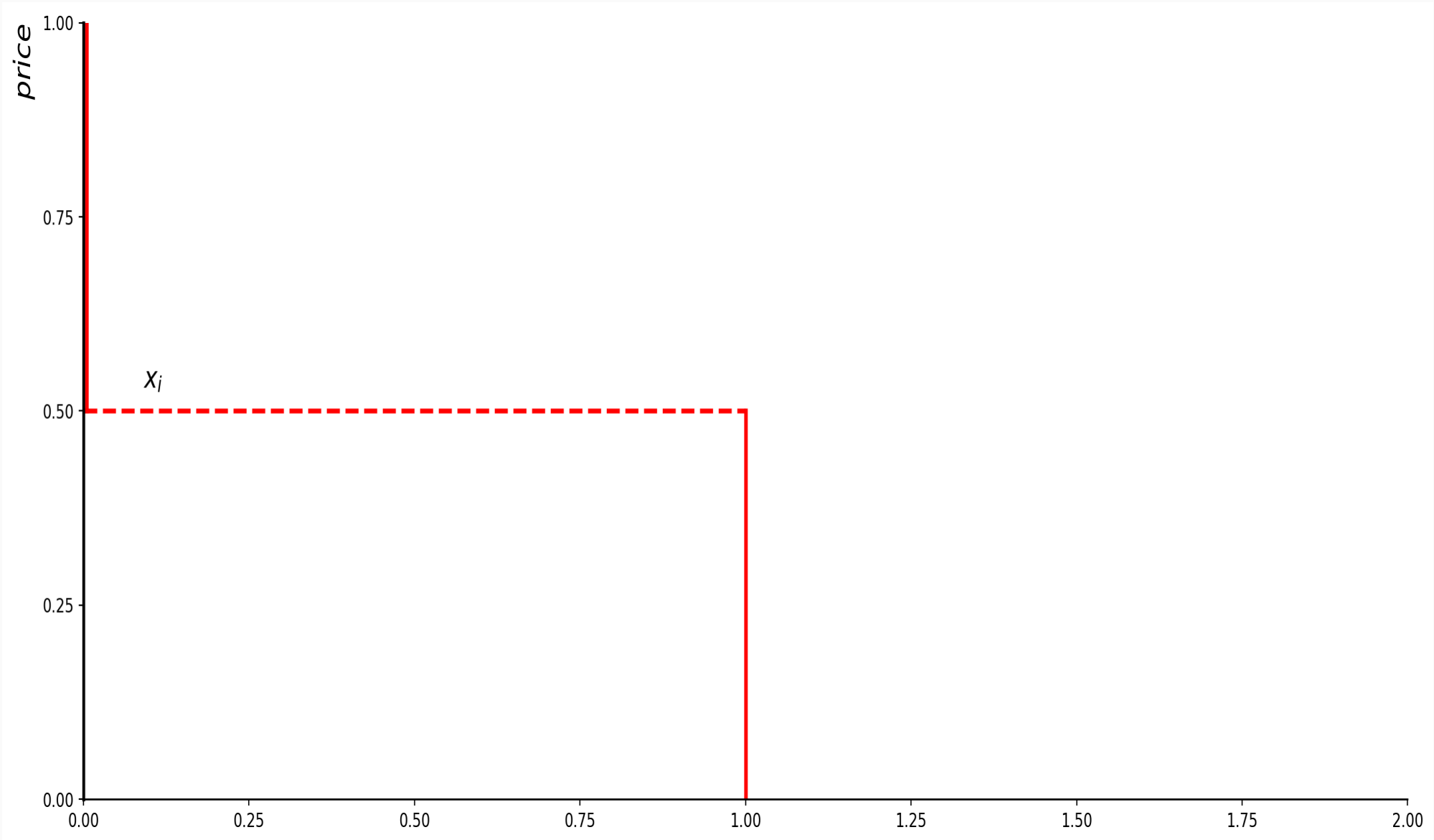
Four basic auctions and the IPVM

Independent private-values model (IPVM)

- A seller of a single, indivisible object; I potential buyers indexed by i .
- Private values: i *privately* observes her valuation $x_i \in X_i$. $X_i = [0, 1]$ is a normalization.
- Independent valuations, $\{x_i\}_i^I$ are independently distributed.
- Symmetry (SIPVM). $\{x_i\}_i^I$ are identically distributed, with a strictly positive, density function $f > 0$. (*)
- Preferences: i observes x_i , gets object with probability q_i , and pays m_i :

$$\begin{aligned} V_i(x, q, m) &= V_i(x_1, \dots, x_I, q_1, q_2, \dots, q_I, m_1, \dots, m_I) \\ &= x_i \cdot q_i - m_i. \end{aligned}$$

At most demand for a single unit



The tool: Bayesian games

A Bayesian game $\Gamma = \{(X_i, \mathcal{X}_i, \mu_i), A_i, V_i\}_{i=1}^I$ is defined by

- I is the number of players.
- $(X_i, \mathcal{X}_i, \mu_i)$ is a probability space that represents player i 's private information.
- A_i is player i 's action space.
- $V_i : \prod X_i \times \prod A_i \rightarrow \mathbb{R}$
indicates player i 's payoffs given type and action profiles respectively.

Solution Concepts

Notation

- $x = (x_1, \dots, x_I)$, $x_{-i} = (x_1, \dots, x_{i-1}, x_{i+1}, x_I)$.
- $X_1, X_2, \dots, X_I$ sets, $X = \prod_{i=1}^I X_i$.
- First order statistic: $x^{(1)} = \max \{x_i\}_{i=1}^I$.
- Second order statistic: $x^{(2)}$.

Second-price, sealed-bid auction (SPA)

Theorem 1 (SPA) IPVM. The strategy profile $\{b_i(x_i) = x_i\}_{i=1}^I$ is a weakly dominant-strategy equilibrium (WDSE) of the SPA.

Proof. Let b_{-i} be the vector of i 's opponents bids.

$$u_i \left(x_i, b_i, b_{-i}^{(1)} \right) = \begin{cases} 0 & \text{if } b_i < b_{-i}^{(1)} \\ \left(x_i - b_{-i}^{(1)} \right) & \text{if } b_i > b_{-i}^{(1)} \\ \left(x_i - b_{-i}^{(1)} \right) P(\text{tie}) & \text{if } b_i = b_{-i}^{(1)} \end{cases}$$

- $x_i < b_{-i}^{(1)} \implies i's$ best response is to lose, thus $b_i = x_i < b_{-i}^{(1)}$.
- $x_i > b_{-i}^{(1)} \implies i's$ best response is to win, thus $b_i = x_i > b_{-i}^{(1)}$.
- $x_i = b_{-i}^{(1)} \implies i$ is indifferent, thus $b_i = x_i$. QED

SPA, remarks

- Weakly dominant strategy equilibrium (WDSE) means a strong recommendation.

Principle: winner's payment does not depend on winner's bid.

- Result is stronger than WDSE of the implicit game. (Continuum technicality.)
- Where is independence used? And symmetry?
- Are there other BNE?
- Solution is efficient.

English auction

Theorem 2 (English auction, clock format) IPVM. The profile $\{b_i(x_i) = x_i\}_{i=1}^I$ is a weakly dominant strategy equilibrium of the EA.

- The proof is straightforward.
- Experimental evidence.

First-price, sealed-bid auction (FPA)

How to bid?

- Bidder i 's expected payoff if value is x_i and bid is b_i

$$= P(\text{Win}) \times (x_i - b_i) + P(\text{Not Win}) \times 0$$

where *Win* means that the bidder gets the object.

- Then one might expect,
 - $b_i < x_i$, the bid is lower than the valuation.
 - Unlikely to have a dominant strategy: how much less? Depends on beliefs about b_{-i} .
 - FPA might not be efficient. (False)

Theorem 3 (First-price, sealed-bid auction (FPA)) Symmetric IPV. The strategy profile

$$\left\{ b_i(x_i) = \int_0^{x_i} y \frac{(I-1)F(y)^{I-2}f(y)}{F(x_i)^{I-1}} dy \right\}_{i=1}^I$$

is a BNE of the FPA. It is the unique symmetric BNE; it is the unique equilibrium in pure strategies.

See Maskin and Riley (2000), Maskin and Riley (2003), Lebrun (1996), Lizzeri and Persico (2000).

- $F(x_i)^{I-1}$ is the cdf of $x_{-i}^{(1)}$, FOS for $(I-1)$ i.i.d. variables.
- $(I-1)F(y)^{I-2}f(y)$ is the pdf of FOS $x_{-i}^{(1)}$.

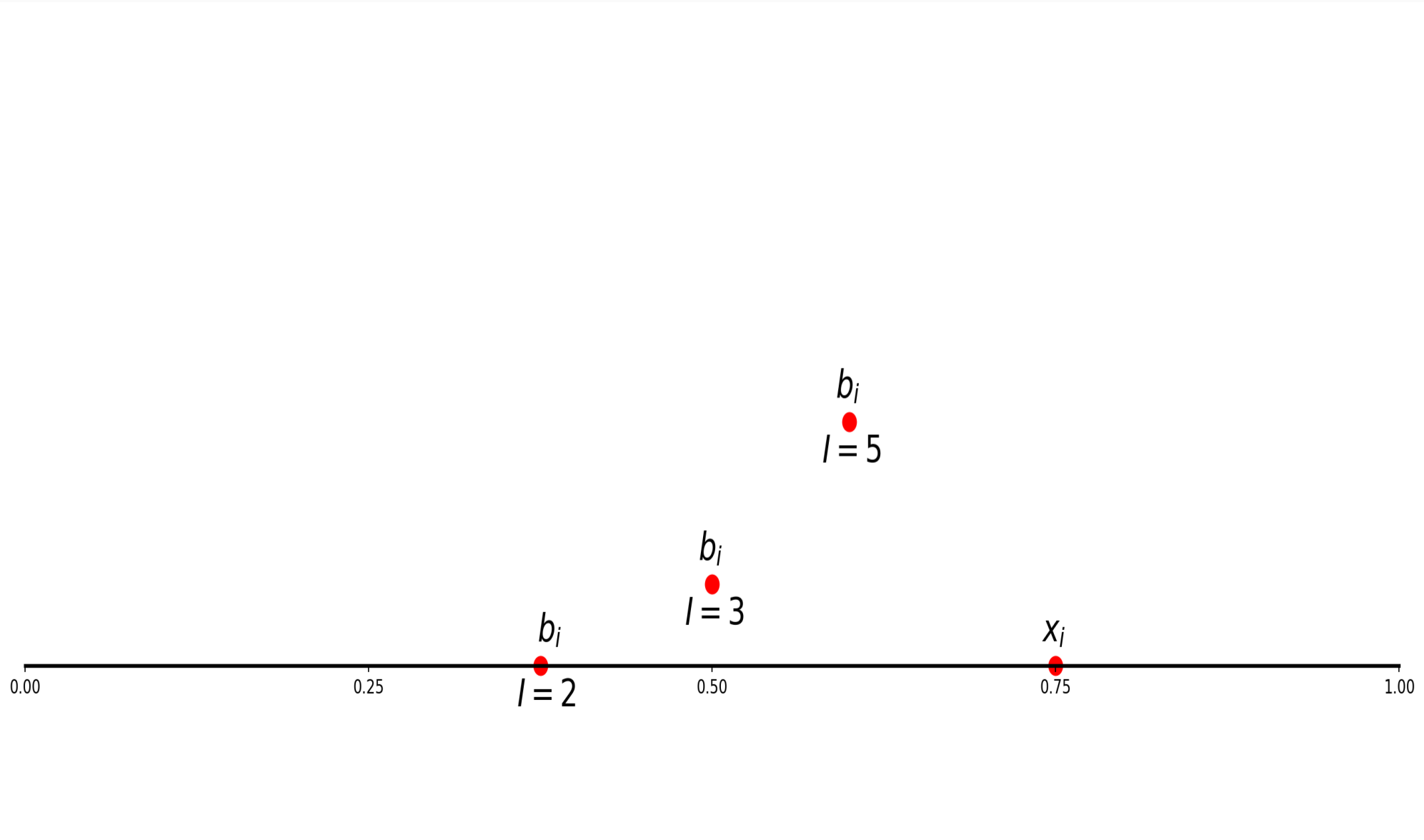
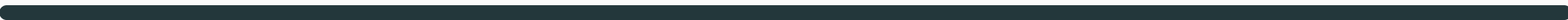
Lemma 1 Alternative formulations of $b(x_i)$ in FPA:

1. $b_i(x_i) = E \left[x_{-i}^{(1)} | x_{-i}^{(1)} \leq x_i \right].$

2. $b_i(x_i) = x_i - \int_0^{x_i} \frac{F(y)^{I-1}}{F(x_i)^{I-1}} dy$

- $F(x_i)^{I-1}$ is cdf of $x_{-i}^{(1)}$, FOS for $(I - 1)$ i.i.d. variables.
- Example $(\forall i) F(x_i) = x_i$, the uniform on $[0, 1]$:

$$b_i(x_i) = x_i - \frac{x_i - 0}{I}.$$



Dutch Auction (DA)

- In its clock format, strategically equivalent to FPA.
- $b_i(x_i) =$ price at which to stop the clock

Aside:

- DA dynamic version of FPA; EA dynamic version of SPA.
- Ausubel auction (multi-unit) dynamic version of multi-unit Vickrey auction.

Seller's preferred auction

Theorem 4 (Revenue equivalence) Symmetric IPV. The SPA, EA, FPA, and DA all generate the same seller's expected revenue.

Proof.

$$\begin{aligned} E[\text{Revenue}_{FPA}] &= E \left[b \left(x^{(1)} \right) \right] \\ &= E \left[E \left[x_{-i}^{(1)} \mid x_{-i}^{(1)} \leq x_i, x_i = x^{(1)} \right] \right] \\ &= E \left[E \left[x^{(2)} \mid x^{(1)} = x_i \right] \right] \\ &= E \left[x^{(2)} \right] \\ &= E[\text{Revenue}_{SPA}] \end{aligned}$$

QED

- See Myerson (1981), Vickrey (1961).

References

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